

PERIVALLON — Master Study Guide

Multispectral VHR satellite detection and risk-classification of illegal waste

Politecnico di Milano · PERIVALLON thesis · Alessandro Potenza

2026-06-28

0. How to use this guide

1. The research question and why it matters
2. The problem framed as risk classification
3. Sensors and the band ladder
4. Material → feature → band (the centerpiece table)
5. Datasets you can actually use
6. The science you must be able to defend
 - 6.1 Object vs material (the honest caveat)
 - 6.2 Mixed pixels set the ceiling
 - 6.3 Chemistry vs texture — the experiment that makes MS defensible
 - 6.4 Generalization (“generalizzazione”) — the advisor’s central claim, quantified
 - 6.5 Remote estimation of asbestos degradation (fills the ID gap)
7. Experimental design (condensed)
8. The risk chain (end to end)
9. Key numbers to memorize
10. Professor Q&A (study for the defence)
11. Glossary

PART II — Papers to study

How to study these papers

(a) Baseline & waste detection

- Deep-learning pipeline for solid waste detection (Gibellini 2025)
- AerialWaste dataset (Torres & Fraternali 2023)
- Global dumpsite detection (Sun et al. 2023)
- CascadeDumpNet (Zhang & Ma 2024)
- CWLD construction-waste landfill dataset (Wang et al. 2024)
- C&D debris volume from drones (2022)
- Disaitek + Airbus Pléiades Neo tracker (2024)
- AerialWaste lightweight/ensemble benchmark (Sharmily et al. 2025)
- fCLIPSeg post-hurricane debris (2025)
- Large-area UAV solid-waste monitoring (Liu et al. 2024)
- Tisza waste detection & change (Magyar et al. 2023)
- Heavy-duty truck detection from satellite (2025)
- Well pads & storage tanks (Ramachandran et al. 2024)
- YOLOv7-OT storage tanks (2024)
- Hybrid-YOLOv5 non-ferrous metals in ELVs (2025)

(b) Asbestos & roof material

- Asbestos rooftops from EnMAP (Shepherd et al. 2025)
- Python AC-roof mapping workflow (Bonifazi et al. 2026)
- AC roofs + weathering from MIVIS (Cilia et al. 2015)
- Asbestos from WV-3 VNIR, 8 bands (Saba et al. 2026)
- Multi-temporal asbestos change detection (Abbasi et al. 2024)
- Asbestos slate drone training-data model (2023)

(c) Plastics & spectral signatures

- USGS Spectral Library v7 / splib07a (Kokaly et al. 2017)
- Knowledge-based aliphatic/aromatic plastic classifier (Zhou et al. 2021)
- Macroplastics mapping with WV-3 SWIR (Aguilar et al. 2025)
- Normalized Difference Plastic Index / NDPI (Guo & Li 2020)
- Global plastic detection with EMIT (Estrela et al. 2025)
- UV-SWIR plastic spectral library (Knaeps et al. 2020)
- MARIDA marine-debris benchmark (Kikaki et al. 2022)
- Cross-domain UAV-SWIR plastic segmentation (2026)

UAV + IoT river-plastic detection (2025)

(d) Sensors, datasets & ablation

- WV-3 VNIR vs SWIR greenhouse ablation (Aguilar et al. 2021)
- Critical wavelengths for C&D waste (Vitek et al. 2025)
- SpectralWaste dataset (Casao et al. 2024)

(e) Foundation models & methods

- DOFA — wavelength-conditioned foundation model (Xiong et al. 2024)
- SatMAE (Cong et al. 2022)
- SpectralGPT (Hong et al. 2024)
- Prithvi-EO-2.0 (Szwarcman et al. 2024)
- SSL4EO-S12 (Wang et al. 2023)
- SoftCon (Wang et al. 2024)
- AnySat (Astruc et al. 2025)
- Resizing & normalization matters (Corley et al. 2024)
- DEFLECT — PEFT for new channels (Thoreau et al. 2025)

(f) Generalization, surveys & context

- PoliMi waste-detection survey (Fraternali et al. 2024)
- FM for Remote Sensing survey (2024)
- Waste RS survey — object-vs-material framing (2024)
- AI-drone waste-management review (2025)
- Vehicle detection with domain adaptation (Koga et al. 2020)

Master table — all 47 papers

0. How to use this guide

This is the single document to study. It has two parts:

- **Part I — Concepts** (this part): the argument, the sensors, the science, the experiment, the numbers, and a professor-defence Q&A. Study this to *understand and be able to explain* the thesis.
- **Part II — Papers**: a study card per paper (47), grouped by theme, with the one number and the one limitation to remember each. Study this to *cite the literature efficiently*.

Suggested study order (≈3 h): §1–§2 (big picture, 20 min) → §3–§4 (sensors + the material↔band table, 35 min) → §5 (datasets, 25 min) → §6 (the science, 35 min) → §7–§8 (experiment + risk chain, 30 min) → §9–§10 (numbers + Q&A, 25 min) → Part II papers (browse cards by tier, 30 min+).

Terminology discipline (advisor's preference): say *generalizzazione* (not “OOD”); *classificare per rischio* (not just *rilevare*); *multiband* (not “spectral cube”); never translate **GSD**. Define EO / MS / HSI / VNIR / SWIR on first use.

The thesis in one sentence. *Detection of illegal-waste sites from very-high-resolution satellite imagery is already solved on RGB; the open, operationally useful problem is to classify those sites by risk — which requires identifying the material — and the thesis measures, honestly and with controlled ablation, how much multiband (VNIR, and where available SWIR) imagery adds over RGB for that material discrimination at VHR.*

1. The research question and why it matters

Central research question. *What is the added value of multiband over RGB for discriminating the materials that determine the hazard of an illegal-waste site, at VHR — and where does it stop helping?*

Three things make this the right question:

1. **Detection is solved, characterisation is not.** RGB models localise dumpsites from shape and context at ~92% F1 (Gibellini 2025). They cannot tell hazardous from inert material. The operational bottleneck for ARPA is not finding sites but *ranking* them.
2. **Risk is defined by material.** Following the environmental-epidemiology formulation, $\text{risk} = \text{hazard} \times \text{exposure} \times \text{magnitude}$ (Fazzo et al. 2020). The *hazard* term is the material (asbestos vs inert rubble), legally codified by the European Waste Catalogue (asbestos = code 17 06 05*). Two of the three risk factors are material-dependent — exactly what RGB cannot see.
3. **The value is real but bounded.** Material identity lives in spectral absorptions outside the visible (NIR/SWIR). But at VHR the spectrum is coarsely sampled and pixels are mixed, so the gain must be *measured*, not assumed — and it differs by material.

Operational context. The thesis sits in PERIVALLON (Horizon Europe, Grant 101073952), with ARPA Lombardia as the end user. The deliverable that matters is a *ranked intervention list*, i.e. decision support, not a detector.

2. The problem framed as risk classification

$\text{risk} = \text{hazard}(\text{material}) \times \text{exposure}(\text{receptors}) \times \text{magnitude}(\text{area})$

- **Hazard** — the material. Legally codified (European List of Waste / EWC; asterisk = hazardous; asbestos 17 06 05*). This is the term MS imagery can supply and RGB cannot.
- **Exposure** — proximity to people, water, agriculture. Computable from GIS layers (out of the imagery, but cheap to add).
- **Magnitude** — site area / quantity. Computable from the detected footprint.

Italian operational anchor. ARPA does not act on detection alone: the *Piano Regionale Amianto* (PRAL) and the *Indice di Degrado* (d.d.g. 13237/2008, thresholds 25 / 44 / 45) already turn material condition into intervention priority. The thesis's risk-tiering therefore mirrors how the agency actually works — a strong “operational relevance” argument. **Fazzo et al. 2020** (the Campania “Land of Fires” GIS risk index) is the citable methodological template.

The contribution junction. Prior art either *detects* sites (RGB, material-blind) or *ranks known sites* by GIS exposure with the hazard supplied from ground records. No published pipeline derives the hazard term *from multispectral imagery* and feeds it to a risk tier. That junction is the novelty.

3. Sensors and the band ladder

The whole thesis turns on which diagnostic absorption falls into which band.

Sensor	VNIR	SWIR	Pan	Role
WorldView-3	8 bands @ 1.24 m	8 bands @ 3.7 m	0.31 m	full ladder incl. R3 chemistry
Pléiades Neo	6 bands @ 1.2 m	none	0.30 m	cross-sensor axis; VNIR-only
SuperDove	8 bands @ 3 m	none	—	free, near-daily; R2 full-VNIR
Sentinel-2	10–20 m	20 m	—	too coarse for material
EnMAP / PRISMA	hyperspectral ~6.5 nm	yes	—	resolves the 2.3 μm triplet but at 30 m

WV-3 VNIR centres (μm): Coastal 0.425 · Blue 0.48 · Green 0.545 · Yellow 0.605 · Red 0.66 · RedEdge 0.725 · NIR1 0.833 · NIR2 0.95 (Yellow + NIR2 unique vs Pléiades Neo). **WV-3 SWIR centres (μm):** 1.21 · 1.57 · 1.66 · **1.73** · 2.16 · 2.20 · 2.26 · **2.33**. **Pléiades Neo VNIR (μm):** DeepBlue 0.425 · Blue 0.483 · Green 0.562 · Red 0.655 · RedEdge 0.725 · NIR 0.84 (no SWIR).

The band ladder (the independent variable of the experiment):

R0 RGB → R1 +RedEdge/NIR → R2 full VNIR (8-band) → R3 +SWIR (WV-3 only).

R3 is the only rung that adds diagnostic chemistry. The Pléiades-Neo asymmetry (it stops at R2) is itself a finding, not a defect.

Data access (no Thomas needed to know this). Both WV-3 and Pléiades Neo are free to a PoliMi/ESA researcher via **ESA Third Party Missions** (project proposal, ~9-week evaluation, 1-year quota). The real risk is **WV-3 SWIR archive availability** over the chosen AOI (SWIR is tasked separately and may be sparse) — not resolution: the old 7.5 m export cap was relaxed, native SWIR is 3.7 m.

4. Material → feature → band (the centerpiece table)

The governing rule: a class is feasible at broadband VNIR if its identity lives in **shape/context**; it is weak if its identity lives in **spectral chemistry**.

Class	Identity carrier	Diagnostic feature (μm)	WV-3 band	On Pléiades Neo	Verdict at VHR
Asbestos-cement	shape + chemistry	chrysotile Mg-OH 2.30-2.33	SWIR8 (2.33)	shape + VNIR weathering proxy	corrugated shape detectable; material confirm needs SWIR
Plastics (polymer)	presence=texture; type=chemistry	C-H 1.215 / 1.730	S1, S4, S8	presence only	presence at VHR; polymer ID needs SWIR
C&D / inert rubble	shape + chemistry	carbonate 2.34; clay 2.20	S8, S6	heap morphology	strong by shape; composition needs SWIR
Foundry slag	chemistry (data-thin)	Fe-oxide 0.87-0.95	NIR1/NIR2; SWIR	weak Fe cue	GAP
Scrap metal	scene=shape; type=chemistry	flat/specular (no feature)	none	scrapyard scene	scene-detectable; type = GAP even with SWIR
Tyres (rubber)	shape / dark target	carbon-black, low R	broadband low	dark piles	piles OK; confuses with shadow/water
Wood / organic	shape (stacks)	cellulose ~2.1/2.3	NIR1/SWIR	stacked piles	good by shape
Sludge	context (lagoon)	water+organic (no sharp feature)	broadband+NIR	lagoon context	GAP (composition)
Big-bags / FIBC	shape but small	woven PP C-H 1.73 (sub-pixel)	S4 (diluted)	shape, tiny	near-GAP
Vehicles / ELV	shape	—	—	shape	excellent (object)
Tanks / cisterns	shape (circular)	—	—	shape	excellent — not a gap
Containers / skips	shape	—	—	shape	strong (object)
Bulky items	shape	—	—	shape	good (object)

Read this in one breath: 8/13 classes are shape-identifiable at 30-50 cm; **5 are chemistry-bound** (asbestos, plastic-type, foundry slag, sludge, scrap composition) — exactly where RGB fails and the MS/risk argument must live. **Asbestos and plastics are the ideal test beds** because a shape cue and a chemistry cue coexist.

The SWIR-8 bottleneck (say this before the prof does). Chrysotile (~2.32), carbonate (~2.34) and plastics C-H (~2.31) all crowd into the *single* WV-3 SWIR8 (~2.33 μm, ~40 nm wide) band. WV-3 cannot *resolve* the triplet within it — it can only flag “something absorbs near 2.33”. Discrimination then leans on the shoulders (2.16/2.20/2.26; 1.21/1.73) plus VNIR shape. A hyperspectral sensor (~6.5 nm) would resolve it; WV-3 won’t, and that limit must be stated.

5. Datasets you can actually use

The structural gap (the thesis opportunity): *no public dataset combines VHR satellite imagery + terrestrial waste + material-level labels.* Every resource sits on one side of the hole.

The bridge: take geolocated waste/asbestos locations and re-image them with WV-3 / Pléiades Neo — this creates the missing multiband material dataset.

Datasets by **role**:

- **Label sources (the actual training data).**
 - *Lombardy asbestos WFS* (Mappatura_2020, 10,903 roofs; Mappature_precedenti, 50,131; EPSG:32632) — **public + geolocated → immediately usable** for the asbestos pilot.
 - *AerialWaste v3* (3,478 waste positives in Lombardy, 22 material categories) — **coordinates withheld (ARPA)** → re-imaging needs an in-group agreement.
 - *DroneWaste* (PERIVALLON UAV, 20 material types, RGB) — material taxonomy; not satellite.
 - *CWLD* (Beijing C&D, GF-2/GE pixel masks) — transfer for the rubble class.
- **Public benchmarks (method dev / band ablation).** EuroSAT MS (the controlled band-ablation testbed, already local); Toulouse Hyperspectral (32 urban materials, VNIR+SWIR, public); GRSS Houston / Pavia (urban-material HSI); SpectralWaste / RECONMATIC / TecNALIA (material-evidence proof); MARIDA/MADOS (material-at-satellite, but marine).
- **Spectral references (justify bands, build endmembers).** USGS splib07a (WV-3 convolutions, in `spectral/`); WaRM (corrugated asbestos-cement); KLUM; MADLib; EnMAP asbestos field library.
- **Pretraining / FM weights.** DOFA (ingests WV-3/PNeo bands natively); SSL4EO-S12 (S2 weights); Prithvi-EO-2.0 (has SWIR); SpectralEarth.

Verified Lombardy auxiliary layers (all EPSG:32632, ArcGIS REST, CC-BY 4.0 — co-registerable with the imagery, no reprojection): asbestos WFS (hard positives; **layer 1** = Mappatura 2020, **layer 4** = temporal status codes 1/2/3/4); DUSAF 7.0 land-use (classes 131 Cave / 132 Discariche / 133 Cantieri / 134 Aree degradate — filter on the `DESCR` string, the numeric code is not queryable); PSC-AGISCO contaminated-site points (status + activity type incl. “abbandono rifiuti”, “discariche abusive”). **The Indice di Degrado is NOT in the WFS** — it must be estimated remotely (see §6).

6. The science you must be able to defend

6.1 Object vs material (the honest caveat)

At VHR, **detection** is largely solved by shape/context (Gibellini F1 92.02%; AerialWaste ensemble 92.41%; Disaitek ~95% on objects $\geq 2 \text{ m}^2$). So the MS value proposition is **not** detection — it is **material/risk classification**, which the VHR literature leaves empty.

6.2 Mixed pixels set the ceiling

Pixel footprints: WV-3 VNIR $\approx 1.5 \text{ m}^2$, SuperDove $\approx 9 \text{ m}^2$, WV-3 SWIR $\approx 14 \text{ m}^2$. Waste objects are decimetre-scale, so **essentially no pixel is pure** — each integrates waste + soil + shadow + vegetation. Lab/library numbers are *upper bounds*; on-satellite numbers are a realistic *mixed-pixel lower bound*. Unmixing is out of experimental scope but governs the ceiling, and the cited target detectors (Aguilar matched-filter, EMIT) are *already* sub-pixel.

6.3 Chemistry vs texture — the experiment that makes MS defensible

A CNN has a receptive field, so “MS-CNN beats RGB-CNN” could be *texture in extra channels*, not chemistry. Use a four-cell design:

	RGB bands	MS bands
Spatial-context-free (per-pixel / per-object mean)	A	B
Full CNN (spatial receptive field)	C	D

- **$B - A$ = pure chemistry/spectral gain** (no spatial info to either) — the number that attributes MS value to *material*.
- **$D - C$ = total MS gain in the deployed CNN** (chemistry + MS-texture).
- **$B - A \approx 0$ but $D - C > 0 \Rightarrow$ the gain is texture** — an honest negative result for the material claim.

Precedent that the context-free baseline works: Saba 2026 Fine-KNN on WV-3 VNIR pixel features reaches Macro-F1 97.6% for asbestos with no spatial kernel.

6.4 Generalization (“generalizzazione”) — the advisor’s central claim, quantified

Mode	Magnitude
Cross-region (same sensor)	–5.1% F1 (Gibellini 92.0→86.9; GR 85.4 / SE 83.8 / RO 91.5)
Cross-sensor (band mismatch)	2–4× error increase (no overlap); 5–25% drop (superset) — GeoCrossBench
Cross-time/season	task-specific double-digit
Atmospheric/illumination	removed by surface-reflectance correction

Does multiband narrow or widen the gap? Genuinely mixed — which is why it is publishable. It **narrows** the gap for cross-region/cross-time on a *fixed* sensor (physical absorptions are more invariant than colour/texture) but **widens** it cross-sensor (the extra bands are exactly where band-centre/width/SNR/calibration mismatch lives; RGB is the lowest common denominator and ports trivially). *Defensible claim*: MS value is conditional on closing the radiometric/band gap; reported naïvely, MS can look worse OOD than RGB. The clean novel experiment is a **2-D table: ablation level (R0→R3) × generalization axis**, reporting the ID–OOD gap per cell. Cheapest highest-ROI fix first: **resize + per-channel normalization** (Corley 2024), then surface reflectance, then wavelength-conditioning (DOFA).

6.5 Remote estimation of asbestos degradation (fills the ID gap)

The Indice di Degrado $ID = (A + \dots + H) \times I$ is a site-inspection score and is **not** in the WFS. Of its factors, only **D (fibre exposure)** has a spectral handle (SWIR chrysotile, weak at WV-3), **B (large cracks)** is partly visible at pan 0.3 m, **A and C are invisible**; exposure factors E-H and age I are GIS/registry (H = ≤ 300 m to sensitive sites is pure GIS). VNIR carries the *biological* weathering signal (moss/lichen: chlorophyll 0.68 + red-edge 0.74); SWIR carries the *matrix-loss/fibre-exposure* signal. Published indices (Valdelamar Martínez 2024, 149 roofs): SWIR chrysotile indices (ISDCAP $\sim 70.6\%$, R2327 $\sim 71.0\%$) beat the VNIR vegetation index ($\sim 40\%$) for *grading*, while VNIR alone is strong for *detection* ($\sim 96\%$ OA). **Proposal:** a two-tier Remote Degradation Proxy — an ordinal spectral surface class (low/med/high) $\times$ a GIS exposure overlay — presented as a *screening pre-filter*, not a substitute for on-site ID. The remote estimator is itself a contribution.

7. Experimental design (condensed)

Three-axis ablation. (A) spectral content R0→R3; (B) sensor: WV-3 (with SWIR) vs Pléiades Neo (VNIR-only) on the same sites; (C) native MS vs pansharpened — classify on **native** MS, use the 0.3 m pan for texture only (pansharpening injects space, not spectrum; use MRA/MTF methods, never GS/IHS; report SAM/ERGAS/Q2n). Run the ladder on **both** the per-pixel row and the full-CNN row (§6.3).

Tasks. T1 binary detection (floor); T2 material multi-class → coarse risk groups; T2-risk risk tier; **T3 asbestos pilot — run first and in full** (the only sub-problem with public geolocated labels coincident with the imagery and a textbook SWIR diagnostic; de-risks the whole pipeline before AerialWaste coordinates are ever needed).

Backbones (run all so the conclusion isn't backbone-specific). Swin-T + RSP extended (weight-inflation / late-fusion / random-init); DOFA (wavelength-conditioned = the structurally-fair spine); CNN-from-scratch (unbiased info-content control); SpectralFormer (spectra-only); per-pixel RF/SVM (the mandatory context-free baseline); Prithvi at R3 only (SWIR specialist). Tooling: TorchGeo, TerraTorch, rasterio, geopandas.

Metrics. macro-F1 **and per-class F1** (MS value concentrates in 1–2 hard materials, washed out by OA); for risk tiers add quadratic-weighted κ and ordinal MAE; headline = **$\Delta F1$ vs R0** with bootstrap 95% CI + paired significance (McNemar).

Splits — make generalization a reported axis. Spatially-blocked ID split (never random tiles); cross-region (hold out provinces); cross-time; cross-sensor (WV-3↔PNeo). Report the ID–OOD gap per band cell.

8. The risk chain (end to end)

RGB detects site (shape/context) → multiband infers material (chemistry) → material → EWC hazard code (asbestos 17 06 05*) → risk = hazard × exposure × magnitude → ranked intervention list for ARPA (report precision@k of high-risk sites, not only global F1). Template: Fazzo 2020; condition term via the remote degradation proxy aligned to the Indice di Degrado thresholds (25/44/45).

9. Key numbers to memorize

- Detection on RGB: Gibellini **F1 92.02%** / **Acc 94.56%**; AerialWaste ensemble 92.41%; Disaitek ~95% ($\geq 2 \text{ m}^2$, vendor).
- Cross-region drop: **-5.1% F1** (92.0→86.9).
- Cross-sensor: **2-4x** error (no band overlap), **5-25%** (superset) — GeoCrossBench.
- Asbestos, VNIR-only: **Saba Macro-F1 97.6%** (WV-3 VNIR, per-pixel). EnMAP ACE 91.4% / 86% field match (Shepherd; 30 m, out of VHR scope).
- WV-3 ablation (plastic-film analogy): OA **90.85 (VNIR) → 96.79 (SWIR) → 97.38 (all)** (Aguilar 2021).
- “Few bands suffice”: **RGB + 2 NIR bands $\approx$ full 768-band HSI** (Vitek/CDW 2025).
- AC-degradation indices (Valdelamar Martínez 2024, 149 roofs): SWIR R2327 ~71% > VNIR ISDVeg ~41% for grading.
- Pixel footprints: WV-3 VNIR ~1.5 m² · SuperDove ~9 m² · WV-3 SWIR ~14 m².
- Indice di Degrado thresholds: **≤ 25 / 26-44 / ≥ 45** ; asbestos WFS = 10,903 roofs (layer 1).
- Diagnostic features: chrysotile **2.30-2.33 μm** ; plastics C-H **1.215 / 1.730 μm** ; carbonate **2.34 μm** .

Do NOT quote as fact (unverified / corrected): Aguilar “+14% kappa” (use OA only; plastic-film/NDPI, not generic material); Saba “HS 97.3% vs MS 74.4%” (paywalled); Bonifazi AC F1 0.87 (screening-tool figure); Cilia “86/89%” precise figures. CascadeDumpNet = **Zhang & Ma 2024** (not Marrocco). Shepherd 2nd author = **Sagi**.

10. Professor Q&A (study for the defence)

Q. What is the task, exactly? Two sub-problems: detection (solved on RGB, F1 92%) and *classification by risk* of detected sites via the material. The contribution is the second, measured by a band ablation.

Q. If RGB detects at 92%, why multiband? Detection $\neq$ risk. RGB wins on morphology, not chemistry; ARPA must prioritise by hazard = material; ~ 5 of 13 classes are chemistry-bound where RGB is blind.

Q. Why not Sentinel-2 (free, has SWIR)? GSD. S-2 is 10–20 m; a dump is 1–4 mixed pixels there. Every VNIR material success is at 0.25–2.5 m, none ≥ 3 m.

Q. What does SWIR buy, and what does Pléiades Neo lose? SWIR holds the molecular overtones (asbestos 2.32, plastics 1.215/1.73, carbonate 2.34). Pléiades Neo (no SWIR) loses direct material confirmation $\rightarrow$ asbestos/plastics collapse to shape + VNIR proxy.

Q. Is WV-3 SWIR useful or too coarse? Honestly partial, possibly weak: 3.7 m + ~ 40 nm bands + atmospheric water attenuate the narrow Mg-OH feature. Measuring this (R3 vs R2) *is* the contribution. A null (R3 $\approx$ R2) is still publishable.

Q. Mixed pixels — doesn't that kill the material claim? It sets the ceiling, not zero. Lab numbers are upper bounds; on-satellite are mixed-pixel lower bounds; the cited detectors are already sub-pixel.

Q. How do you prove the MS gain is chemistry, not texture? The four-cell design: B–A (per-pixel MS–RGB) isolates pure spectral gain; B–A ≈ 0 with D–C > 0 means the gain was texture (honest negative).

Q. Does multiband help or hurt generalization? Narrows within-sensor (physics $>$ appearance), widens cross-sensor (band mismatch) unless reflectance is harmonised — a publishable 2-D table.

Q. Can you predict the Indice di Degrado remotely? Not fully — only surface factors A–D, and only D weakly via SWIR; exposure E–H and age I are GIS. Output an ordinal screening class, not a fake continuous ID.

Q. Is the contribution novel? The 2-D ablation $\times$ generalization table at the empty VHR material operating point, the chemistry-vs-texture decomposition, the material $\rightarrow$ EWC $\rightarrow$ risk chain, and the asbestos pilot demonstrator — the combination, hedged.

Q. Is the data feasible? Free via ESA TPM (~ 9 wk); asbestos pilot has public WFS labels now; the real risk is WV-3 SWIR archive over the AOI and SWIR co-registration at 3.7 m.

Q. Why not hyperspectral (EnMAP/PRISMA)? They resolve the 2.3 μm triplet but at 30 m — useless over decimetre waste. No sensor today has both hyperspectral resolution and sub-metre GSD over wide areas; the thesis works in that realistic gap.

Q. Biggest risk of a null result? That WV-3 SWIR at 3.7 m is too diluted to beat VNIR for asbestos (R3 $\approx$ R2). Still a valid, publishable finding; the four-cell design guards against the opposite error.

11. Glossary

- **EO** Earth Observation. **MS** multispectral (few broad bands). **HSI** hyperspectral (100s of narrow bands). **VNIR** visible–near-infrared (0.4–1.0 μm). **SWIR** short-wave infrared (1.0–2.5 μm ; molecular overtones).
 - **GSD** Ground Sampling Distance (pixel size on the ground; keep untranslated). **Pan** panchromatic (one broad high-resolution band).
 - **SAM** / **ACE** / **SID** spectral matching: Spectral Angle Mapper / Adaptive Cosine (Coherence) Estimator / Spectral Information Divergence. **ERGAS** / **Q2n** pansharpening quality metrics.
 - **EWC European Waste Catalogue** / **European Waste Codes** (List of Waste, Decision 2000/532/EC; asbestos 17 06 05). (Not the ML term “Elastic Weight Consolidation”).*
 - **Endmember** pure-material reference spectrum. **Mixed pixel** a pixel integrating several materials. **Unmixing** estimating sub-pixel abundances. **NDPI** Normalized Difference Plastic Index. **RSP** Remote Sensing Pretraining.
 - **Indice di Degrado** Lombardy AC-roof site-inspection degradation score (d.d.g. 13237/2008). **Generalizzazione** performance on data unlike training (cross-region / cross-sensor / cross-time); **ID–OOD gap** in-domain minus out-of-domain score.
 - **DOFA** wavelength-conditioned foundation model (variable band sets). **C-H** / **Mg-OH** / **CO₃** molecular bonds whose SWIR absorptions identify plastics / asbestos / carbonate.
-

PART II — Papers to study

(Study cards for all 47 papers follow, grouped by theme, each with the one number and the one limitation to remember. Citations reconciled against the verified bibliography.)

How to study these papers

Read for the **thesis argument**, not for completeness. The spine is one question: *what is the added value of multispectral (MS) over RGB for classifying illegal waste by material/risk at VHR?* Every paper is either evidence for, evidence against, a method, or context for that question.

- **Tiers.** Spend most time on **critical** papers (the baseline you extend, the datasets you use, the asbestos anchors, the spectral library, DOFA). **High** papers are supporting evidence you must cite by number. **Medium** papers are breadth — know the one-line role and move on.
- **For each paper extract four things:** (1) the *one headline number* (F1/OA/mAP + GSD + #bands); (2) the *method in a sentence*; (3) the *honest limit* (where exam questions live); (4) *where it sits* in the MS-vs-RGB argument.
- **Recurring through-line:** few well-chosen bands beat both RGB and full hyperspectral (cdw-2025, spectralwaste-2024, aguilas-2021); SWIR carries the C-H polymer / Mg-OH asbestos diagnostics (zhou-2021, plastics-uv-swir, emit-2025); but a no-SWIR VHR sensor reframes the work as a generalization / VNIR-sufficiency problem (saba-2026, abbasi-2024).

(a) Baseline & waste detection

Deep-learning pipeline for solid waste detection (Gibellini 2025)

- **Cite:** Gibellini, Fraternali, Boracchi, Morandini, Martinoli, Diecidue & Malegori 2025, *Waste Management Bulletin*. DOI 10.1016/j.wmb.2025.100246 · arXiv 2502.06607
- **Relevance:** critical · **Tags:** baseline, swin-t, rsp, aerialwaste
- **What it is:** End-to-end RGB pipeline (tile → Swin-T classifier → Grad-CAM → GIS) co-designed with ARPA Lombardia.
- **Why it matters:** This *is* the baseline the thesis extends to multiband.
- **Key things to remember:**
 - Best model = Swin-T + RSP + 20 cm GSD + 100 m context: **F1 92.02%, Acc 94.56%**; Swin-T > ResNet-50 everywhere.
 - 36-condition factorial: 2 archs × 3 GSD × 3 context × 2 pretrains.
 - Cross-country (Greece/Sweden/Romania) mean F1 86.92% (−5.10 pp).
 - Utility: +63.2% sites found, −60.2% area inspected at threshold 0.7.
- **Honest limit:** RGB-only — no spectral value, cannot distinguish materials; RSP gives only modest gain because its pretraining GSD mismatches the 20 cm task.

AerialWaste dataset (Torres & Fraternali 2023)

- **Cite:** Torres & Fraternali 2023, *Scientific Data* 10:63. DOI 10.1038/s41597-023-01976-9 — real title “*AerialWaste dataset for landfill discovery in aerial and satellite images*” (corrected)
- **Relevance:** critical · **Tags:** dataset, aerialwaste, rgb
- **What it is:** First public dataset for illegal-landfill discovery; 10,434 images (v3 ~11.7k), 3 sources (AGEA 20 cm, WV-3 30 cm, Google Earth 50 cm).
- **Why it matters:** The main training dataset; defines the 22 material categories and class imbalance.
- **Key things to remember:**
 - Three annotation levels: binary scene, 22 Type-of-Object + 7 Storage-Mode labels, 169 images with 841 masks.
 - Baseline ResNet50+FPN: 87.99% AP, 80.70% F1 (94.5% AP on AGEA-only).
 - neg:pos = 2:1; built with ARPA over 487 comuni; **only 11 tiles tagged “corrugated sheets / presumed asbestos.”**
- **Honest limit:** RGB-only across 3 mismatched GSDs — cannot itself test MS-vs-RGB; coordinates withheld; authors list “add NIR” as future work.

Global dumpsite detection (Sun et al. 2023)

- **Cite:** Sun et al. 2023, *Nature Communications* 14:1444. DOI 10.1038/s41467-023-37136-1
- **Relevance:** high · **Tags:** waste, global, dumpsites
- **What it is:** First global fine-grained dumpsite dataset (~2,500 sites, 7 countries, 0.3–1 m) + BCA-Net.
- **Why it matters:** Shows DL scales globally; quantifies operational savings — motivates MS as the next step.
- **Key things to remember:** 4 dumpsite types; mean sensitivity 98.0%, precision ~70.1%; 6 days vs 6 months (–96.8% time); Global Dumpsite Index correlates with GNI.
- **Honest limit:** ~70% precision because morphology, not material, is seen; China over-represented.

CascadeDumpNet (Zhang & Ma 2024)

- **Cite:** Zhang & Ma 2024, *Remote Sensing of Environment* 313:114349. DOI 10.1016/j.rse.2024.114349 — **two authors, NOT Marrocco/“et al.”** (corrected)
- **Relevance:** high · **Tags:** dumpsite, pléiades, object-detection, automl
- **What it is:** Dual-stage CNN detection + AutoML on Pléiades 0.5 m with a Context-Fusion module.
- **Why it matters:** Newest VHR-satellite dumpsite comparator; false-alarm-suppression idea is transferable.
- **Key things to remember:** **84.6% mAP**; transfers Shenzhen → Shanghai/Guangzhou.
- **Honest limit:** Pléiades commercial VHR; object-level, no spectroscopy.

CWLD construction-waste landfill dataset (Wang et al. 2024)

- **Cite:** Wang et al. 2024, *Scientific Data* 11. DOI 10.1038/s41597-024-03240-0
- **Relevance:** high · **Tags:** construction, dataset, gaofen, segmentation
- **What it is:** Pixel-level C&D-landfill segmentation dataset (3,653 samples) from GF-2 ~80 cm + Google Earth ~50 cm.
- **Why it matters:** Provides the per-pixel masks AerialWaste lacks for the rubble/C&D class.
- **Key things to remember:** Improved DeepLabV3+: **F1 88.9% / IoU 82%**; two Beijing districts.
- **Honest limit:** Single-region; C&D only; RGB.

C&D debris volume from drones (2022)

- **Cite:** *Drones* 6(10):279, 2022. DOI 10.3390/drones6100279
- **Relevance:** high · **Tags:** debris, drone, fcn, volume
- **What it is:** Drone + SfM 3D reconstruction + FCN segmentation of C&D debris with volume estimation.
- **Why it matters:** Drone route for rubble with 3D volume — a capability 2D satellite detectors lack.
- **Key things to remember:** IoU 0.9 for concrete debris; adds volume, not just footprint.
- **Honest limit:** UAV close-range; RGB.

Disaitek + Airbus Pléiades Neo tracker (2024)

- **Cite:** Disaitek / Airbus / CNES 2024 — vendor case study (grey literature)
- **Relevance:** high · **Tags:** operational, pléiades-neo, commercial
- **What it is:** Operational service on Pléiades / Pléiades Neo 30 cm detecting waste ≥ 2 m².
- **Why it matters:** Strongest “someone does this operationally from satellite”; market relevance.
- **Key things to remember:** ~95% reliability claim; qualifies type (ELV, construction, tyres, vegetal); CNES/Airbus agreement 27 May 2024.
- **Honest limit:** Vendor claim, no peer review or published method.

AerialWaste lightweight/ensemble benchmark (Sharmily et al. 2025)

- **Cite:** Sharmily et al. 2025, arXiv 2508.18315 (PARTIAL — arXiv-only)
- **Relevance:** medium · **Tags:** aerialwaste, ensemble, benchmark
- **What it is:** Lightweight CNN + transformer ensembles on AerialWaste.
- **Why it matters:** Shows the field still does binary scene classification.
- **Key things to remember:** **Binary F1 92.41%**; no public move yet to per-material classification.
- **Honest limit:** Binary only; preprint.

fCLIPSeg post-hurricane debris (2025)

- **Cite:** 2025, arXiv 2504.12542
- **Relevance:** medium · **Tags:** debris, foundation-model, clipseg
- **What it is:** Fine-tuned CLIPSeg for debris segmentation on aerial RGB.
- **Why it matters:** SAM/CLIP-family adaptation recipe for data-thin shape classes.
- **Key things to remember:** Dice 0.70; event-agnostic across three hurricanes.
- **Honest limit:** Aerial RGB, disaster debris (not illegal waste); modest Dice.

Large-area UAV solid-waste monitoring (Liu et al. 2024)

- **Cite:** Liu et al. 2024, *Applied Sciences* 14(5):2084. DOI 10.3390/app14052084
- **Relevance:** medium · **Tags:** drone, segmentation, dual-branch
- **What it is:** Dual-branch semantic segmentation of solid-waste piles over ~450 km² UAV imagery.
- **Why it matters:** Evidence drone work treats waste as one generic “heap” class — the material gap.
- **Key things to remember:** OA >94%, recall 88.6%; no per-material breakdown.
- **Honest limit:** Single generic pile class; UAV.

Tisza waste detection & change (Magyar et al. 2023)

- **Cite:** Magyar, Cserep, Vincellér & Molnár 2023, arXiv 2303.14521
- **Relevance:** medium · **Tags:** change-detection, sentinel-2, planetscope
- **What it is:** RF on Sentinel-2 (13 b) + PlanetScope (4 b, 3 m) for dump hotspots & floating-waste barriers on the Tisza.
- **Why it matters:** Classic ML on MS reaches high accuracy; NIR is the key band — but only for large accumulations.
- **Key things to remember:** ~96% CV accuracy; Plastic Index $PI = NIR/(NIR+Red)$ cardinal; detection floor ~30 m².
- **Honest limit:** At 10–30 m, works only on dense accumulations >100 m²; no spectroscopy.

Heavy-duty truck detection from satellite (2025)

- **Cite:** 2025, PMC12302943 (open access)
- **Relevance:** medium · **Tags:** vehicles, containers, object-detection
- **What it is:** CenterNet + Mask R-CNN ensemble classifying container size/status and trucks.
- **Why it matters:** Per-class anchor (containers/trucks by shape); flags the absent ELV detector.
- **Honest limit:** Objects, not material; no ELV class.

Well pads & storage tanks (Ramachandran et al. 2024)

- **Cite:** Ramachandran, Irvin, Omara, Ng & Jackson 2024, *Nature Communications* 15. DOI 10.1038/s41467-024-50334-9
- **Relevance:** high · **Tags:** tanks, object-detection, large-scale
- **What it is:** DL mapping of >169k storage tanks and >70k well pads across US basins.
- **Why it matters:** SOTA tank detection at sub-metre VHR — comparator for the cistern class.
- **Key things to remember:** Tanks **P 0.962 / R 0.968**; continental scale.
- **Honest limit:** Shape-based; no material discrimination.

YOLOv7-OT storage tanks (2024)

- **Cite:** 2024, *Remote Sensing* 16(23):4510. DOI 10.3390/rs16234510
- **Relevance:** high · **Tags:** tanks, yolo, object-detection
- **What it is:** YOLOv7 + CBAM tank detector with edge re-stitching.
- **Why it matters:** Confirms tanks are a well-served object class, not a gap.
- **Key things to remember:** 90% accuracy / 95.9% precision.
- **Honest limit:** Shape-based; not material/risk.

Hybrid-YOLOv5 non-ferrous metals in ELVs (2025)

- **Cite:** 2025, *Scientific Reports* 15. DOI 10.1038/s41598-025-02683-8
- **Relevance:** medium · **Tags:** scrap, metal, elv, close-range
- **What it is:** Close-range IR detection of copper/aluminium/steel in end-of-life vehicles.
- **Why it matters:** Documents the scrap-composition gap and the object-vs-material boundary.
- **Key things to remember:** 84.2% mAP; **NOT remote sensing** — needs spectrum / close range.
- **Honest limit:** Close-range IR, not VHR satellite.

(b) Asbestos & roof material

Asbestos rooftops from EnMAP (Shepherd et al. 2025)

- **Cite:** Shepherd, **Sagi**, Zagron & Ben-Dor 2025, *Scientific Reports* 15:24166. DOI 10.1038/s41598-025-09738-w (2nd author corrected to Sagi)
- **Relevance:** critical · **Tags:** enmap, hyperspectral, sam, satellite
- **What it is:** Spaceborne HSI AC-roof mapping (EnMAP 230 bands, 30 m) with field-spectra calibration + 8-classifier cascade.
- **Why it matters:** The most direct satellite justification for the asbestos pilot; the HSI upper bound.
- **Key things to remember:** **86% positive-match** vs GT; best single classifier ACE 91.4% OA, $\kappa=0.87$; 2,714 field spectra; “6-of-8” consensus.
- **Honest limit:** 30 m → mixed-pixel errors (rubble/paint mimic asbestos); calibrated to semi-arid Negev; heavy manual curation; **30 m is out of VHR scope.**

Python AC-roof mapping workflow (Bonifazi et al. 2026)

- **Cite:** Bonifazi, Aurigemma, Salas-Cáceres, Lorenzo-Navarro, Serranti, Paglietti, Bellagamba & Malinconico 2026, *Geomatics* 6(3):41. DOI 10.3390/geomatics6030041
- **Relevance:** critical · **Tags:** wv-3, vnir, python, open-source, multi-temporal
- **What it is:** Open-source Python workflow (Py6S → MLC → building aggregation) for AC detection + temporal monitoring on WV-3 over Mantova.
- **Why it matters:** Most recent, reproducible, directly comparable; the technical stack to reuse.
- **Key things to remember:** 16 WV-3 bands (VNIR 1.24 m + SWIR 3.7 m → resampled 1 m); **F1 0.87 at 30% threshold**; multi-temporal caught 46.6% of 133 registered removals.
- **Honest limit:** A *screening* tool (needs field verification); thresholds Mantova-specific.

AC roofs + weathering from MIVIS (Cilia et al. 2015)

- **Cite:** Cilia, Panigada, Rossini, Candiani, Pepe & Colombo 2015, *ISPRS Int. J. Geo-Information* 4(2):928-941. DOI 10.3390/ijgi4020928
- **Relevance:** high · **Tags:** hyperspectral, mivis, sam, weathering, italian
- **What it is:** Airborne MIVIS (102 bands, 3 m) SAM classification of AC roofs in Brianza + a deterioration index (ISD).
- **Why it matters:** Foundational Italian AC reference; SAM + endmember + deterioration-index workflow + cadastre integration.
- **Key things to remember:** 2-step SAM PA 89% / UA 86%; 3,170 AC roofs; ISD built on chlorophyll (moss/lichen), not fibre release.
- **Honest limit:** ISD is a vegetal-colonisation proxy, not fibre hazard; 3 m excludes roofs <36 m²; airborne not scalable.

Asbestos from WV-3 VNIR, 8 bands (Saba et al. 2026)

- **Cite:** Saba et al. 2026, *Journal of Hazardous Materials* 508:141864. DOI 10.1016/j.jhazmat.2026.141864
- **Relevance:** high · **Tags:** wv-3, vnir, classification
- **What it is:** 32 classifiers on WV-3's 8 VNIR bands for AC-roof classification.
- **Why it matters:** Strongest evidence that **VNIR-only suffices** for asbestos at fine VHR — central to the no-SWIR framing.
- **Key things to remember:** Fine-KNN Macro-F1 97.6%; binary AC ~99–100%; red-edge & NIR drive it; GSD 1.24 m.
- **Honest limit:** WV-3 VNIR at 1.24 m (not SuperDove 3 m); classifier zoo, single region.

Multi-temporal asbestos change detection (Abbasi et al. 2024)

- **Cite:** Abbasi et al. 2024, *Remote Sensing Applications: Society and Environment*. DOI 10.1016/j.rsase.2024.101167
- **Relevance:** high · **Tags:** aerial, obia, deep-learning, no-swir
- **What it is:** Nearmap aerial VHR (no SWIR) + DenseNet121 + LSTM/Conv1D multi-temporal AC detection.
- **Why it matters:** Counter-evidence that asbestos is detectable at fine VHR by shape + temporal cues *without SWIR*.
- **Key things to remember:** OA 95.8–96.0%, AC 94%; temporal stacking compensates for missing bands.
- **Honest limit:** Needs multi-temporal aerial; RGB-only; no spectral proof.

Asbestos slate drone training-data model (2023)

- **Cite:** 2023, PMC10575463 (open access)
- **Relevance:** medium · **Tags:** drone, rgb, training-data
- **What it is:** A DL training-data model from drone RGB of asbestos slate roofs.
- **Why it matters:** Drone-RGB asbestos route; documents the annotation bottleneck.
- **Honest limit:** UAV RGB, no spectral confirmation; annotation-heavy.

(c) Plastics & spectral signatures

USGS Spectral Library v7 / splib07a (Kokaly et al. 2017)

- **Cite:** Kokaly et al. 2017, *USGS Data Series 1035*. Report DOI 10.3133/ds1035 · data DOI 10.5066/F7RR1WDJ
- **Relevance:** critical · **Tags:** signature-library, usgs
- **What it is:** The world reference spectral library: ~2,400 signatures, 350–2500 nm, 2,151 channels, with sensor-convolved subsets (S2, WV-3).
- **Why it matters:** Source for the spectral-signature figures and endmembers; the lab-to-sensor bridge.
- **Key things to remember:** Covers polymers, AC/chrysotile, construction materials; local copy in `spectral/`; no-data sentinel — **1.23e+34**; convolved S2 (13 b) & WV-3 (16 b) subsets.
- **Honest limit:** Ideal lab conditions — real scenes need atmospheric correction, BRDF, continuum removal, mixed-pixel handling.

Knowledge-based aliphatic/aromatic plastic classifier (Zhou et al. 2021)

- **Cite:** Zhou, Kuester, Bochow, Bohn, Brell & Kaufmann 2021, *Remote Sensing of Environment*. DOI 10.1016/j.rse.2021.112598
- **Relevance:** high · **Tags:** wv-3, swir, aliphatic-aromatic, classifier
- **What it is:** Training-free decision-tree classifier using C-H features in WV-3's 8 narrow SWIR bands.
- **Why it matters:** Explains *why narrow SWIR (WV-3) resolves PE vs PET and broad S2 B11/B12 does not* — the band-width argument.
- **Key things to remember:** 3 polymer clusters; producer accuracy >80% aliphatic-vs-aromatic; validated lab → airborne → WV-3.
- **Honest limit:** Thresholds need per-sensor re-calibration; does not transfer to no-SWIR sensors; weak field truth.

Macroplastics mapping with WV-3 SWIR (Aguilar et al. 2025)

- **Cite:** Aguilar, Sousa, Uhrin, Gudino-Elizondo & Biggs 2025, *Environmental Monitoring and Assessment*. DOI 10.1007/s10661-025-14125-z
- **Relevance:** high · **Tags:** wv-3, swir, plastic, high-resolution
- **What it is:** Matched-filter mapping of terrestrial macroplastics with WV-3 SWIR (8 b, 3.7 m) + lab reflectance.
- **Why it matters:** Lab signatures transfer to satellite in the SWIR; WV-3 as the material gold standard at VHR.
- **Key things to remember:** Precision 92.5% (95.3% high-confidence); lab-to-image $r=0.95$; smallest detected 80–150 m².
- **Honest limit:** Matched filter favours large homogeneous instances; “polymer-coated” paints indistinguishable from plastic; arid-context only.

Normalized Difference Plastic Index / NDPI (Guo & Li 2020)

- **Cite:** Guo & Li 2020, *ISPRS J. Photogrammetry & Remote Sensing*. DOI 10.1016/j.isprsjprs.2020.09.009
- **Relevance:** high · **Tags:** wv-3, swir, spectral-index, ndpi
- **What it is:** Defines NDPI, a SWIR band-ratio index on WV-3 for urban plastic quantification.
- **Why it matters:** Origin of the single most predictive feature in WV-3 plastic ablations.
- **Key things to remember:** NDPI uses C-H absorptions (~1.57/1.73 μm); top single feature (drives the 96.79% SWIR-only OA in Aguilar 2021).
- **Honest limit:** Closed-access (details *unverified*); SWIR-dependent, no transfer to SuperDove.

Global plastic detection with EMIT (Estrela et al. 2025)

- **Cite:** Estrela et al. 2025, *Geophysical Research Letters* 52:e2024GL112416. DOI 10.1029/2024GL112416
- **Relevance:** high · **Tags:** plastic, hyperspectral, emit, global-scale
- **What it is:** First continental-scale HDPE/PVC detection from orbit (EMIT, 285 b, 60 m) via SWIR matched filtering.
- **Why it matters:** Confirms polymer signatures are orbit-detectable *when SWIR is available*.
- **Key things to remember:** Absorption centres 1215/1417/1732/2313 nm; gypsum FP-rejection at 2200 nm; hotspots Spain/Sicily/Mexico/Taiwan.
- **Honest limit:** 60 m → mostly sub-pixel; low-contrast polymers under-represented; buried landfills missed.

UV-SWIR plastic spectral library (Knaeps et al. 2020)

- **Cite:** Knaeps et al. 2020, *Earth System Science Data* 12:77. DOI 10.5194/essd-12-77-2020
- **Relevance:** high · **Tags:** plastic, hyperspectral, uv-swir, signature-library
- **What it is:** Open ASD FieldSpec library of virgin/marine/washed-ashore plastics, 350–2500 nm.
- **Why it matters:** Physical basis for plastic-detection figures; endmembers for matching/mixing.
- **Key things to remember:** 8 diagnostic absorptions; **1215 and 1732 nm** most robust; wetting attenuates SWIR up to 90% but preserves shape.
- **Honest limit:** Lab-equivalent, white-plastic bias; submerged plastic collapses NIR/SWIR.

MARIDA marine-debris benchmark (Kikaki et al. 2022)

- **Cite:** Kikaki et al. 2022, *PLOS ONE* 17(1):e0262247. DOI 10.1371/journal.pone.0262247
- **Relevance:** high · **Tags:** marine-debris, sentinel-2, benchmark
- **What it is:** First public pixel-level S2 marine-debris benchmark (1,381 patches, 15 classes).
- **Why it matters:** Most-cited MS waste benchmark; shows SWIR value and the 10 m resolution ceiling.
- **Key things to remember:** RF F1 ~0.79; separability rides on 865 nm + SWIR B11/B12; removing SWIR degrades it.
- **Honest limit:** At 10 m, debris is sub-pixel and confuses with vegetation/foam; learns regional distributions.

Cross-domain UAV-SWIR plastic segmentation (2026)

- **Cite:** 2026, *Remote Sensing* 18(1):182. DOI 10.3390/rs18010182
- **Relevance:** high · **Tags:** drone, swir, hyperspectral, u-net, generalization
- **What it is:** Attention-gated residual U-Net on UAV hyperspectral SWIR (900–1700 nm).
- **Why it matters:** Polymer-type ID needs SWIR; also an OOD/generalization datapoint.
- **Key things to remember:** 96.8% acc / 91.1% F1; driven by 1215 & 1732 nm.
- **Honest limit:** UAV HSI, not satellite; cross-domain still data-bound.

UAV + IoT river-plastic detection (2025)

- **Cite:** 2025, *Journal of Hazardous Materials Advances*. DOI 10.1016/j.hazadv.2025.100348
- **Relevance:** medium · **Tags:** plastic, drone, iot, edge
- **What it is:** On-board edge UAV detector for river plastic (Bogor, Indonesia).
- **Why it matters:** Drone-RGB plastic-presence route.
- **Key things to remember:** 92% accuracy; plastic by appearance, not polymer spectrum.
- **Honest limit:** RGB presence only; UAV.

(d) Sensors, datasets & ablation

WV-3 VNIR vs SWIR greenhouse ablation (Aguilar et al. 2021)

- **Cite:** Aguilar, Jiménez-Lao & Aguilar 2021, *Remote Sensing* 13(11):2133. DOI 10.3390/rs13112133
- **Relevance:** critical · **Tags:** wv-3, vnir, swir, ablation, methodology-canonical
- **What it is:** Three-way OBIA ablation (VNIR / SWIR / All) for plastic-covered greenhouse mapping in Almería.
- **Why it matters:** The canonical “spectral added value” template — the exact framing the thesis adopts.
- **Key things to remember:** **OA 90.85 (VNIR) / 96.79 (SWIR) / 97.38 (All)**; NDPI top SWIR feature; 16 WV-3 bands.
- **Honest limit:** Material is *plastic*, not asbestos — transferable methodology, not an AC substitute; whitewash masks the SWIR signature.

Critical wavelengths for C&D waste (Vitek et al. 2025)

- **Cite:** Vitek, Zbiral & Nezerka 2025, *Resources, Conservation & Recycling*. DOI 10.1016/j.resconrec.2025.108123 · arXiv 2501.02239
- **Relevance:** medium · **Tags:** construction-demolition, wavelengths, band-selection
- **What it is:** HSI (768 bands, 400–1000 nm) band-selection over 10 C&D materials.
- **Why it matters:** The “few bands suffice” argument — motivation for MS over both RGB and full HSI.
- **Key things to remember:** **RGB + 2 NIR bands $\approx$ full 768-band HSI** (0.96 vs 0.87 RGB-only); optimal extra bands $\sim$ 650–750 + $\sim$ 850–1000 nm; whole-spectrum features *degrade* the model.
- **Honest limit:** Controlled lab, clean samples — needs re-calibration for outdoor/satellite; concrete stays hardest.

SpectralWaste dataset (Casao et al. 2024)

- **Cite:** Casao, Peña, Sabater et al. 2024, arXiv 2403.18033
- **Relevance:** medium · **Tags:** dataset, hyperspectral, sorting, multimodal
- **What it is:** First RGB + HSI (224 bands, 900–1700 nm) dataset from a real sorting plant, with SAM label transfer.
- **Why it matters:** RGB+spectral fusion beats either modality — supports late fusion.
- **Key things to remember:** CMX RGB+HSI mIoU 58.2% vs 48.4% RGB vs 54.3% HSI; HSI wins big on thin classes.
- **Honest limit:** Controlled conveyor; SWIR stops at 1700 nm; 6 imbalanced classes.

(e) Foundation models & methods

DOFA — wavelength-conditioned foundation model (Xiong et al. 2024)

- **Cite:** Xiong et al. 2024, *DOFA: Neural Plasticity-Inspired Multimodal Foundation Model for EO*, arXiv 2403.15356
- **Relevance:** critical · **Tags:** dofa, cross-modal, any-bands
- **What it is:** A dynamic hypernetwork conditioned on each band's centre wavelength that generates patch-embedding weights — sensor-agnostic.
- **Why it matters:** **SuperDove/WV-3/PNeo-ready** (feed centre wavelengths, no retraining) — the structurally-fair ablation spine.
- **Key things to remember:** Pretrained MIM on 5 modalities (S1/S2/Gaofen/NAIP/EnMAP); Base 86M / Large 300M; in TorchGeo.
- **Honest limit:** High-res pretraining thin (NAIP 1 m RGB only); SuperDove's Coastal/Yellow/RedEdge may be under-represented; not a guaranteed cross-sensor fix.

SatMAE (Cong et al. 2022)

- **Cite:** Cong et al. 2022, NeurIPS, arXiv 2207.08051
- **Relevance:** high · **Tags:** multispectral, pretraining, mae
- **What it is:** MAE adapted to satellite via band grouping + spectral positional encoding.
- **Why it matters:** Historical baseline; quantifies RGB→MS transfer gain.
- **Key things to remember:** +14% transfer on EuroSAT vs ImageNet; EuroSAT 98.98%.
- **Honest limit:** Rigid grouping needs per-sensor re-adapting; 10/20 m only.

SpectralGPT (Hong et al. 2024)

- **Cite:** Hong et al. 2024, IEEE TPAMI 46(8):5227–5244. DOI 10.1109/TPAMI.2024.3362475 · arXiv 2311.07113
- **Relevance:** high · **Tags:** spectral, 3d-transformer, foundation-model
- **What it is:** Spectral FM with 3D spectral masking, up to 600M params.
- **Why it matters:** Opposite philosophy to DOFA — deep 3D modelling vs sensor flexibility.
- **Honest limit:** Locked to S2 12-band; 8-band SuperDove needs redesign + re-pretrain; tested only on land-cover.

Prithvi-EO-2.0 (Szwarcman et al. 2024)

- **Cite:** Szwarcman et al. 2024, arXiv 2412.02732
- **Relevance:** high · **Tags:** nasa-ibm, multi-temporal, mae
- **What it is:** 3D MAE on 4.2M HLS samples at 30 m, 6 fixed bands incl. SWIR1/SWIR2; ViT-L/H.
- **Why it matters:** Backbone with *native* SWIR and multi-temporal support.
- **Key things to remember:** 75.6% mean on GEO-Bench (+8 pp vs Prithvi-1.0).
- **Honest limit:** 6 fixed bands; 30 m → 10× GSD gap to SuperDove; weak candidate here.

SSL4EO-S12 (Wang et al. 2023)

- **Cite:** Wang et al. 2023, arXiv 2211.07044
- **Relevance:** high · **Tags:** ssl, sentinel-1-2, dataset
- **What it is:** 3M-patch S1/S2 SSL dataset + benchmark across all 13 S2 bands.
- **Why it matters:** Most practical S2 starting point — ready 13-band weights.
- **Key things to remember:** SSL beats ImageNet ~10% in linear probe.
- **Honest limit:** Checkpoint collection, rigid sensors; SuperDove's Yellow 612 nm has no S2 analogue; 3 m unseen.

SoftCon (Wang et al. 2024)

- **Cite:** Wang et al. 2024, arXiv 2405.20462
- **Relevance:** high · **Tags:** contrastive, multilabel, continual-pretrain
- **What it is:** Soft multi-label contrastive continual-pretraining from DINOv2, random-init first layer for mismatched bands.
- **Why it matters:** Proof that “random-init first layer + strong backbone” is a cheap, solid adaptation for 8 bands.
- **Key things to remember:** ViT-B SOTA (86.8 on BigEarthNet-10%); wins 10/11 downstream.
- **Honest limit:** Soft labels noisy; locked to S2 13-band; LULC labels too coarse for material.

AnySat (Astruc et al. 2025)

- **Cite:** Astruc et al. 2025, CVPR 2025, arXiv 2412.14123
- **Relevance:** high · **Tags:** any-resolution, any-modality, jepa
- **What it is:** JEPA multimodal model + scale-adaptive patch encoder for any resolution/scale/modality.
- **Why it matters:** Direct DOFA alternative; SuperDove's 3 m is inside its pretrain range.
- **Honest limit:** Modest pretrain scale; no 8-band VNIR sensor seen → SuperDove stays OOD (fine-tune only).

Resizing & normalization matters (Corley et al. 2024)

- **Cite:** Corley, Robinson, Dodhia, Lavista-Ferres & Najafirad 2024, CVPR-W (PBVS), arXiv 2305.13456
- **Relevance:** high · **Tags:** benchmark, normalization, preprocessing
- **What it is:** Shows preprocessing (resize to 224, channel-wise normalization) often dominates the choice of pretraining.
- **Why it matters:** Mandatory fairness guardrail for the MS-vs-RGB comparison.
- **Key things to remember:** Resize to 224 → +11.16% EuroSAT, +32.28% So2Sat; ImageNet (preprocessed correctly) rivals RS-SSL.
- **Honest limit:** Benchmarks pad extra channels rather than testing real band-config shift; no GSD-shift study.

DEFLECT — PEFT for new channels (Thoreau et al. 2025)

- **Cite:** Thoreau, Marsocci & Derksen 2025, ICCV 2025, arXiv 2503.09493
- **Relevance:** medium · **Tags:** parameter-efficient, multispectral-adapter, peft
- **What it is:** Adapter extending a ViT to new channels with <1% extra params.
- **Why it matters:** Efficient alternative to weight-inflation/random-init if GPU-bound; plugs on DOFA/Prithvi.
- **Key things to remember:** On par with full fine-tune at 5–10× fewer params.
- **Honest limit:** Value depends on the base GFM; needs explicit band indices; not tested on 8-band sensors.

(f) Generalization, surveys & context

PoliMi waste-detection survey (Fraternali et al. 2024)

- **Cite:** Fraternali, Morandini & Herrera González 2024, *Waste Management* 189:88–102. DOI 10.1016/j.wasman.2024.08.003 · arXiv 2402.09066
- **Relevance:** critical · **Tags:** survey, gap-analysis
- **What it is:** Systematic PRISMA survey of 50 RS waste-detection works (1987–2023).
- **Why it matters:** The thesis rationale — names the RGB limitation and material-classification gap.
- **Key things to remember:** 1,235→50 papers; only 11/50 use DL, nearly all RGB; declared gaps: no benchmark, weak generalization, **material identification unaddressed (needs ≤ 30 cm + MS)**, FMs unexplored.
- **Honest limit:** A taxonomy survey, not quantitative.

FM for Remote Sensing survey (2024)

- **Cite:** Xiao et al. 2024/2025, IEEE GRSM, arXiv 2410.16602
- **Relevance:** high · **Tags:** foundation-model, survey
- **What it is:** Taxonomy of 30+ RS foundation models with pretrain-dataset tables.
- **Why it matters:** Related-work scaffold to position DOFA/AnySat (SatMAE→SpectralGPT→DOFA→AnySat).
- **Honest limit:** Coverage stops ~Q3 2024; no unified benchmark; doesn't address 8-band sensors.

Waste RS survey — object-vs-material framing (2024)

- **Cite:** Fraternali, Morandini & Herrera González 2024, *Waste Management* 189:88–102 (published version; object-vs-material angle)
- **Relevance:** high · **Tags:** survey, gap-analysis, object-vs-material
- **What it is:** Catalogues what RS detects vs what is absent.
- **Why it matters:** Independent verification of the per-class gaps and the object-vs-material distinction.
- **Key things to remember:** No detectors for foundry slag, sludge, big bags, scrap composition.
- **Honest limit:** Same paper as the survey above (kept separate for the 13-class cross-check).

AI-drone waste-management review (2025)

- **Cite:** 2025, *Drones* 9(8):550. DOI 10.3390/drones9080550
- **Relevance:** medium · **Tags:** drone, survey, yolo
- **What it is:** PRISMA review of ~10 UAV+DL waste studies.
- **Why it matters:** Cross-check the drone side of the per-class table.
- **Key things to remember:** YOLO dominant (YOLOv8 ~97% acc for dumps); thin per class.
- **Honest limit:** Small corpus; UAV-only; RGB.

Vehicle detection with domain adaptation (Koga et al. 2020)

- **Cite:** Koga, Miyazaki & Shibasaki 2020, *Remote Sensing* 12(3):575. DOI 10.3390/rs12030575
- **Relevance:** medium · **Tags:** vehicles, object-detection, domain-adaptation
- **What it is:** Region-based vehicle detector on 30–50 cm VHR with CORAL + adversarial unsupervised domain adaptation.
- **Why it matters:** Object anchor for vehicles **and** a label-free cross-region generalization technique.
- **Key things to remember:** Domain adaptation gives +10% in the target domain without labels.
- **Honest limit:** RGB shape-based; vehicles, not waste material.

Master table — all 47 papers

Citekey	Year	Relevance	One-line role
gibellini-2025-pipeline	2025	critical	Swin-T+RSP RGB baseline (F1 92.02%) the thesis extends
torres-2023-aerialwaste	2023	critical	AerialWaste — the main RGB training dataset (22 classes)
global-dumpsites-2023	2023	high	Global VHR dumpsite detection + socioeconomic factors
cascadedumpnet-2024	2024	high	Pléiades 0.5 m dual-stage dumpsite detector (84.6% mAP)
cwld-2024	2024	high	Beijing C&D-landfill pixel-level VHR dataset (F1 88.9%)
cd-debris-drone-2022	2022	high	Drone+SfM 3D C&D debris segmentation & volume
disaitek-airbus-pleiades-neo-2024	2024	high	Operational Pléiades Neo waste service (vendor, ~95%)
ramachandran-2024-tanks-wellpads	2024	high	SOTA VHR storage-tank/well-pad detection (P 0.962)
yolov7ot-tanks-2024	2024	high	YOLOv7-OT tank detector (95.9% precision)

aerialwaste-ensemble-2025	2025	medium	Lightweight/ensemble benchmark on AerialWaste (F1 92.41%)
fclipseg-debris-2025	2025	medium	Fine-tuned CLIPSeg for debris (Dice 0.70)
uav-solidwaste-2024	2024	medium	Dual-branch UAV waste-pile segmentation (OA >94%)
tisza-2023-waste-change	2023	medium	RF on S2/PlanetScope river-waste change detection (~96%)
heavyduty-truck-satellite-2025	2025	medium	Satellite truck/container detection; flags ELV gap
hybrid-yolov5-elv-2025	2025	medium	Close-range IR scrap-metal in ELVs; material-gap evidence
shepherd-2025-asbestos-enmap	2025	critical	EnMAP 30 m hyperspectral asbestos (86% match) — HSI upper bound
bonifazi-2026-ac-python	2026	critical	Open-source WV-3 AC mapping workflow (F1 0.87)
cilia-2015-ac-weathering	2015	high	MIVIS airborne AC + deterioration index (PA 89%)
saba-2026-asbestos-wv3-vnir	2026	high	WV-3 VNIR-only asbestos (Macro-F1 97.6%) — no-SWIR evidence
abbasi-2024-asbestos-changedetection	2024	high	Aerial VHR + temporal asbestos without SWIR (OA ~96%)
asbestos-slate-drone-2023	2023	medium	Drone-RGB asbestos slate training-data model
kokaly-2017-splib07a	2017	critical	USGS reference spectral library (350–2500 nm)
zhou-2021-plastic-classifier	2021	high	WV-3 SWIR knowledge-based aliphatic/aromatic plastic classifier
aguilar-2025-macroplastics-wv3	2025	high	WV-3 SWIR matched-filter macroplastics (P 92.5%)
guo-li-2020-ndpi-wv3	2020	high	NDPI spectral index for WV-3 plastic (closed-access)
emit-2025-plastic	2025	high	EMIT 60 m global HDPE/PVC detection from orbit
plastics-uv-swir-2020	2020	high	UV-SWIR plastic spectral library (diagnostic bands)
marida-2022-marine-debris	2022	high	S2 pixel-level marine-debris benchmark (RF F1 ~0.79)
plastic-uav-swir-2026	2026	high	UAV-SWIR plastic segmentation (F1 91.1%) + OOD datapoint
plastic-uav-iot-2025	2025	medium	Edge UAV river-plastic presence detection (92%)
aguilar-2021-wv3-ablation	2021	critical	Canonical VNIR/SWIR/All ablation (OA

			90.85/96.79/97.38)
cdw-2025-critical-wavelengths	2025	medium	“RGB+2 NIR bands $\approx$ full HSI” band-selection for C&D
spectralwaste-2024-dataset	2024	medium	RGB+HSI sorting dataset; fusion > single modality
xiong-2024-dofa	2024	critical	Wavelength-conditioned FM — SuperDove/WV-3/PNeo-ready
cong-2022-satmae	2022	high	MAE + band grouping; historical MS-FM baseline
spectralgpt-2024	2024	high	3D spectral FM (S2-locked, 600M)
szwarcman-2024-prithvi-eo2	2024	high	NASA/IBM 30 m multi-temporal FM with native SWIR
wang-2023-ssl4eo-s12	2023	high	S1/S2 SSL dataset + 13-band pretrained weights
wang-2024-softcon	2024	high	Soft-contrastive continual pretrain; random-init first layer
anysat-2024	2025	high	JEPA any-resolution/modality FM (DOFA alternative)
corley-2024-resizing	2024	high	Preprocessing dominates RS benchmarks — fairness guardrail
thoreau-2025-deflect	2025	medium	PEFT adapter for new channels (<1% params)
fraternali-2024-survey	2024	critical	PoliMi waste survey — names the RGB/material gap
fm-rs-survey-2024	2024	high	Taxonomy of 30+ RS foundation models
waste-rs-survey-2024	2024	high	Published waste survey — object-vs-material gap catalogue
uav-waste-review-2025	2025	medium	PRISMA review of UAV+DL waste studies (YOLO-dominant)
vehicle-detection-da-2020	2020	medium	VHR vehicle detector + unsupervised domain adaptation (+10%)